

Mendel’s Garden of Secrets

II PUC · Ch 4 · Principles of Inheritance and Variation

Starring Bindu 👧 & Prof. BAIO 🦉

A BAIO storybook by Narendra Sir · Biology Lecturer



The Monk’s Garden

Prof. BAIO took Bindu to a pea garden in a monastery. "Here, from 1856 to 1863, **Gregor Mendel** crossed garden peas. He studied seven pairs of contrasting traits — tall and dwarf, round and wrinkled, yellow and green..."

💡 **REMEMBER**

Mendel is called the Father of Genetics.

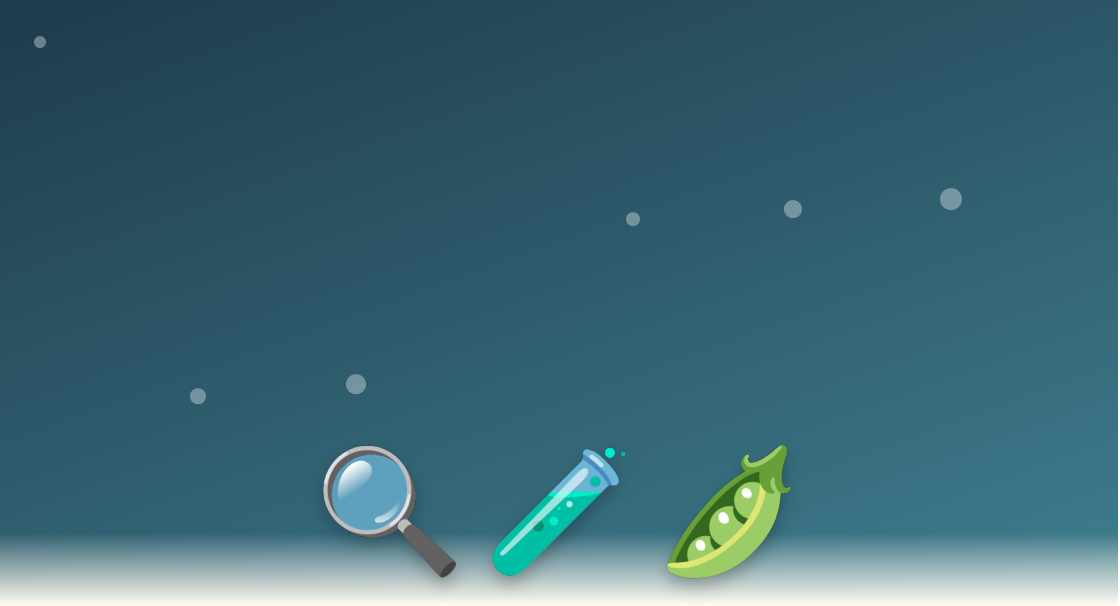


Tall Meets Dwarf

Mendel crossed a tall plant with a dwarf one. Every F_1 plant was tall — tall was **dominant**, dwarf **recessive**. In the F_2, tall and dwarf appeared in a **3 : 1** ratio, and the genotypes TT : Tt : tt in **1 : 2 : 1**. This gave the **law of dominance** and the **law of segregation**.

💡 **REMEMBER**

Law of segregation: the two alleles separate during gamete formation — the law of purity of gametes.



The Test Cross

"How do I know whether a tall plant is TT or Tt?" asked Bindu. "Cross it with the recessive parent, tt — a **test cross**," said the owl. "All tall means TT. Half tall and half dwarf — 1 : 1 — means Tt."

> 💡 **REMEMBER**
> **Monohybrid test cross ratio = 1 : 1.**



Two Traits Together

Next Mendel followed two traits at once — seed shape and colour. The F_2 came in a **9 : 3 : 3 : 1** ratio. The two traits sorted out separately — the **law of independent assortment**.

💡 **REMEMBER**

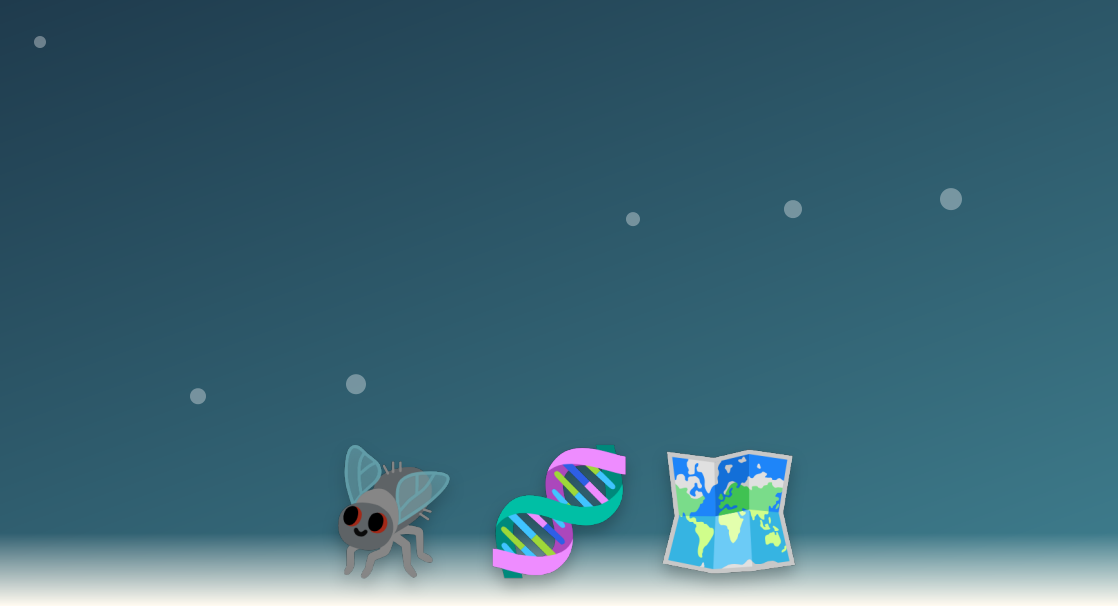
Dihybrid F_2: 9 round yellow : 3 round green : 3 wrinkled yellow : 1 wrinkled green.



When the Rules Bend

In snapdragon, red × white gave pink — **incomplete dominance**, with F_2 in 1 : 2 : 1. In the ABO blood groups, I^A and I^B both showed up together in group AB — **co-dominance**. Three alleles of one gene meant **multiple alleles**.

> 💡 **REMEMBER**
> **ABO blood group: 3 alleles, 6 genotypes, 4 phenotypes.**



Genes on Chromosomes

Sutton and Boveri said genes ride on chromosomes — the **chromosomal theory of inheritance**. **T.H. Morgan** studied fruit flies (Drosophila) and found **linkage** and **recombination**. Sturtevant used recombination frequency to draw gene maps.

💡 **REMEMBER**

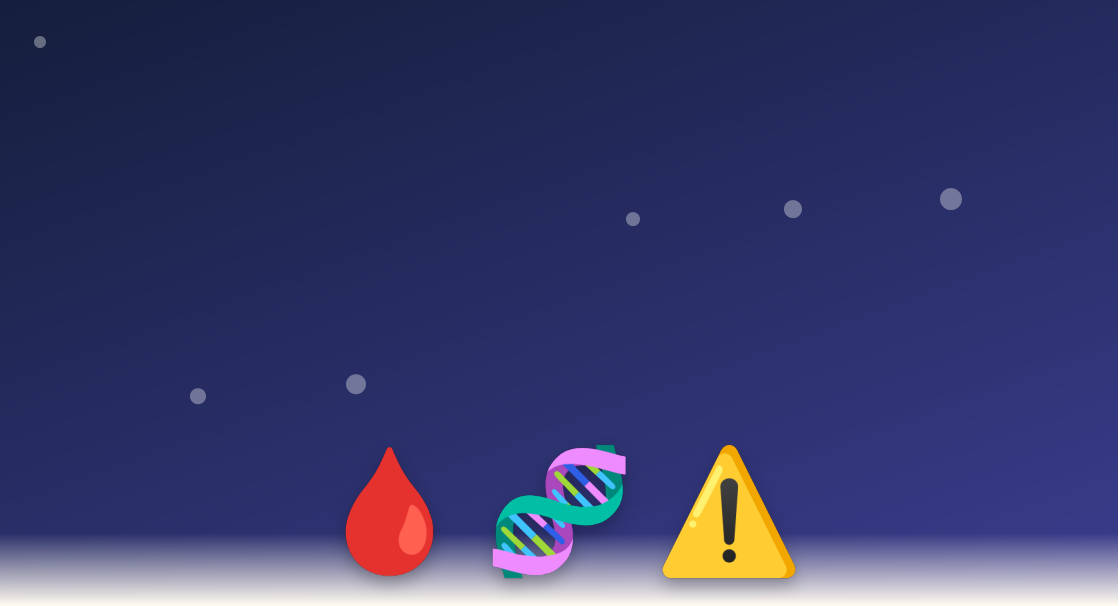
Drosophila: short life cycle (2 weeks), many offspring, easy to grow.



Boy or Girl?

Sex is decided in different ways. Grasshoppers are **XO** type. Humans and Drosophila are **XY** type. In birds the female is ZW — **ZW–ZZ** type. Honey bee males come from unfertilised eggs — **haplodiploidy**.

> 💡 **REMEMBER**
> **In humans, the father's sperm (X or Y) decides the sex of the child.**



When Things Go Wrong

A **mutation** can cause disorders. **Haemophilia** and colour blindness are X-linked recessive. **Sickle-cell anaemia** is autosomal recessive (Glu → Val in β-globin). An extra chromosome 21 causes **Down's syndrome**; XXY causes Klinefelter's and X0 causes Turner's. Doctors trace traits in families using **pedigree analysis**.

💡 **REMEMBER**

Failure of chromatids to segregate causes aneuploidy (gain or loss of a chromosome).



KEY TERMS

Words you met in this story

Tap a word to jump to its page. Can you explain each one?

Gregor Mendel p.1 · dominant p.2 · recessive p.2 · 3 : 1 p.2 · 1 : 2 : 1 p.2 · law of dominance p.2 · law of segregation p.2 · test cross p.3 · 9 : 3 : 3 : 1 p.4 · law of independent assortment p.4 · incomplete dominance p.5 · co-dominance p.5 · multiple alleles p.5 · chromosomal theory of inheritance p.6 · T.H. Morgan p.6 · linkage p.6 · recombination p.6 · XO p.7 · XY p.7 · ZW–ZZ p.7 · haplodiploidy p.7 · mutation p.8 · Haemophilia p.8 · Sickle-cell anaemia p.8 · Down's syndrome p.8 · pedigree analysis p.8



QUICK QUIZ

Do you remember?

1. The monohybrid F_2 phenotypic ratio is:

- 1 : 2 : 1
- 3 : 1
- 9 : 3 : 3 : 1
- 1 : 1

2. Pink snapdragon flowers show:

- Co-dominance
- Incomplete dominance
- Linkage
- Mutation

3. Blood group AB shows:

- Co-dominance
- Incomplete dominance
- Dominance
- Pleiotropy

4. Honey bee males are produced by:

- Fertilised eggs
- Unfertilised eggs
- Budding
- Fission

5. Down's syndrome is caused by trisomy of chromosome:

- 13
- 18
- 21
- X

Answers: 1-3 : 1 · 2-Incomplete dominance · 3-Co-dominance · 4-Unfertilised eggs · 5-21